Kodamarthi Sathwik

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.19
Narayana High School	Class XII (CBSE)	2023	94.20%
Sri Chaitanya Techno School	Class X (SSC)	2021	98.83%

EXPERIENCE

Google : SWE Intern

Automated the regression triaging using statistical correlations to reduce the Mean Time to Resolve

- Designed an agentic system to automate the triaging process of backend regressions critical for Search Latency.
- Collected and joined data across several supporting tables to rule out base-level issues as the cause of a regression.
- Directly impacted Mean Time To Resolve (MTTR) for regressions while reducing manual toil.

Research on IBLTs : Prof Praveen Tammana

- Simulated a LossRadar system subjected to random adversary attacks to evaluate the resilience of IBLTs.
- Evaluated the performance of ListGuard within LossRadar environments under malicious traffic conditions.
- Developed and implemented traffic analysis algorithms utilizing Entropy, Mutual Information, and Cosine Similarity.

PROJECTS

Centralized Communication Library

- Implemented content lifecycle management with version history and rollback support.
- Built a quorum-based review system with decision recording and role-based access control.
- Containerized the full-stack application using Docker Compose for API, web, and PostgreSQL services.

ProbScript: Probabilistic Programming Language

- Designed a DSL for probabilistic programming with native support for distributions and random variables.
- Built a complete compiler frontend using Flex, Bison, semantic analysis, type checking, and symbol resolution.
- Developed a rich type system supporting vectors, matrices, sets, maps, probabilistic types, type inference, and coercion.

VMDT: Verifiable Multipath Decentralized Transport Protocol

- Designed a secure multipath transport protocol using Raft
- Integrated Shamir's Secret Sharing, AES-256-CTR encryption, and peer-to-peer routing.
- Built reliable data transfer over UDP with custom packet formats, acknowledgments, and retransmissions.
- Implemented a decentralized cluster architecture with dynamic share management and k-out-of-n reconstruction.

Cricket Tournament Management System

- Engineered a full-stack tournament management platform with real-time updates, fixtures, scorecards, and player statistics.
- Implemented efficient server-side functions and stored procedures in MySQL to automate tournament operations and statistics.
- Optimized database performance by structuring queries and procedures for scalability and fast data retrieval

RISC-V Processor Simulator

- Engineered a RISC-V simulator supporting assembly, instruction execution, memory management, and debugging.
- Implemented a configurable cache system with write-back/write-through policies and performance metrics.
- Developed instruction pipelining concepts to improve execution efficiency and reduce clock cycles.

gemOS Memory Management

- Implemented the memory management functions for gemOS
- Created mappings between virtual address spaces and page table entries.
- Modified system calls to enable memory access to the allocated VMAs using page table manipulations

PUBLICATIONS

ListGuard: Mitigating Pollution Attacks on In-Network IBLTs

Harish S A, Vignesh S, Sathwik Kodamarthi, et al

Designed a defense that detects and filters adversarially injected entries in programmable-switch IBLTs, preserving decode success under attack. Using realistic traces, we demonstrate attacks and show that ListGuard restores affected statistics with low memory overhead.

SKILLS

- Languages : C, C++, Python, Assembly, SQL
- Web Technologies : HTML, CSS, ReactJS, NextJS, Node.js, ExpressJS
- Databases : MySQL, Postgres, SQLite
- DevOps : Docker, Git, GitHub
- Other Tools : LaTeX, Shell Scripting, Linux, CMake, Flex, Bison

RELEVANT COURSES

- **Mathematics and Statistics** : Probability and Random Variables, Statistics, Linear Algebra, Calculus, Discrete Mathematics
- **Theory & Algorithms** : Data Structures, Algorithms, Approximation Algorithms, Theory of Computation, Compiler Design
- **Systems & Networks** : Operating Systems, Computer Architecture, Computer Networks, Database Systems, Software Engineering
- **Specialized** : Foundations of Machine Learning, Quantum Computing, Quantum Cryptography, Blockchain Technologies

POSITIONS OF RESPONSIBILITY

- Overall Head, Kludge IITH
- Core, Kludge IITH
- Development Core, Lambda IITH
- Web Coordinator, Media Council IITH

EXTRACURRICULAR

- Member of **National Service Scheme (NSS)**
- **Cisco Campus Ambassador** at IIT Hyderabad for the academic year 2024-2025.